Shane Sweetman

Project update meeting
Week 8

A TMD-oriented analysis of $\pi^+\pi^-$ pairs in e^+e^- collisions

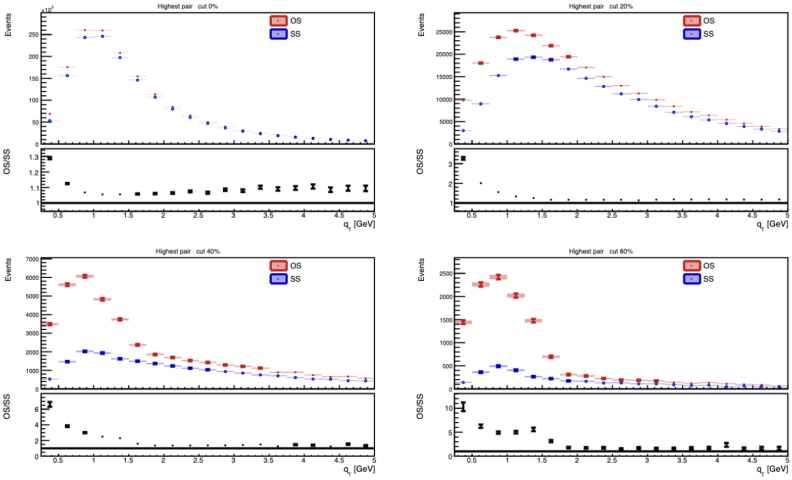
Outline (Week 8)

- Recap of last week.
- Theory code from Valerio and Matteo.
- Shape comparisons
- Interpret the cut dependence in terms of favoured vs unfavoured fragmentation.
- Introduce more compact cut-dependent observables such as $\langle q_T \rangle$ and $q_T^{\max}$.
- Quick confirmation of how Valerio's code works.
- Event shape comparisons.

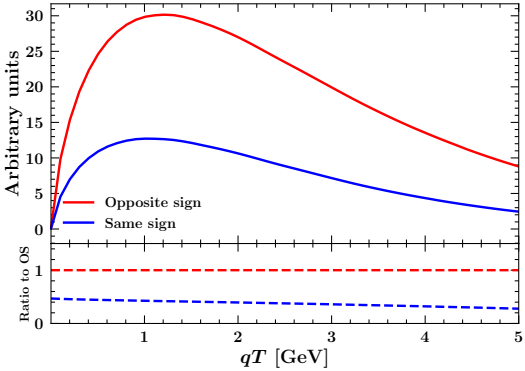
Recap from last week

Main update

Last week the main plotting change was the use of a reduced number of bins to make the overall shape comparisons clearer.

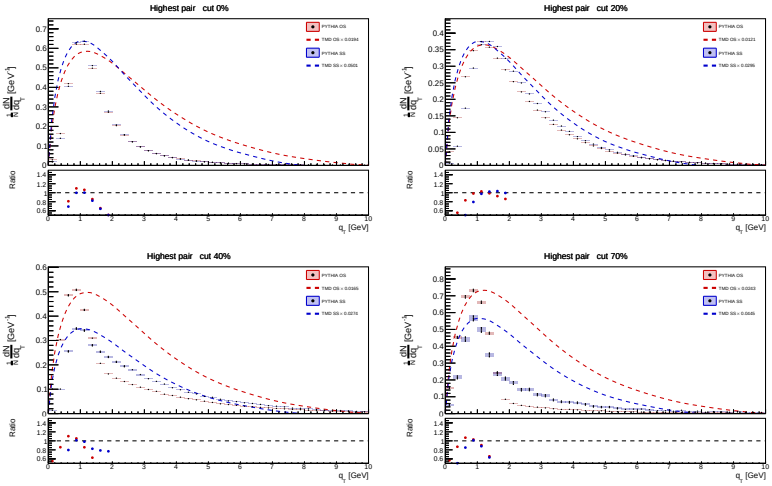


Valerio and Matteo code

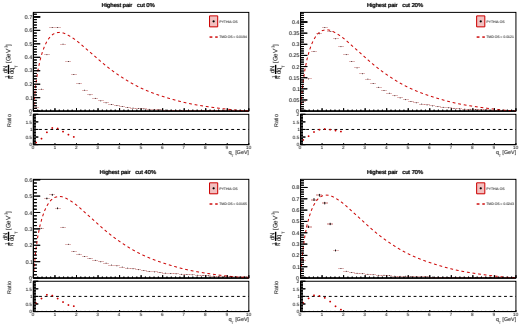


- Updated code now gives both OS and SS theory curves in one run.
- Theory setup is fixed at $z_1 = z_2 = 0.7$ and $Q = M_Z$.
- Most useful as a shape comparison in the small-q_T region.
- Normalisation still has to be rescaled when overlaid with the Pythia distributions.

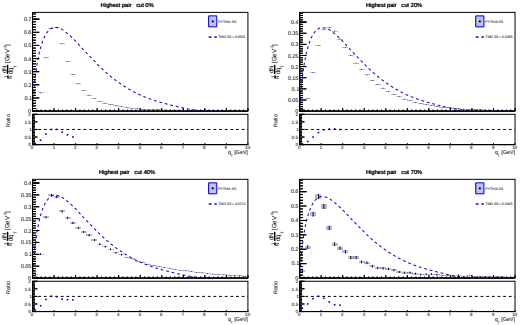
Peak-matched shape comparison



OS and SS theory shapes

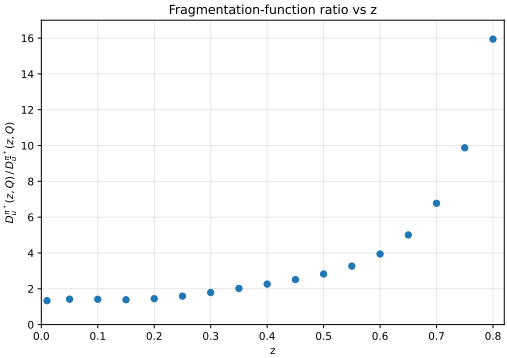


OS shape

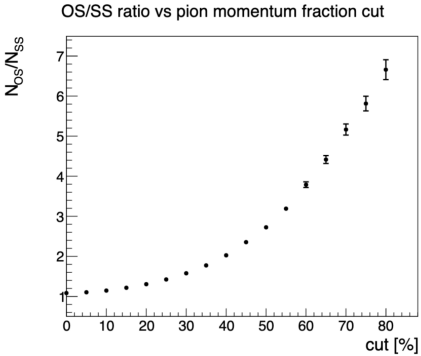


SS shape

Favoured vs unfavoured fragmentation



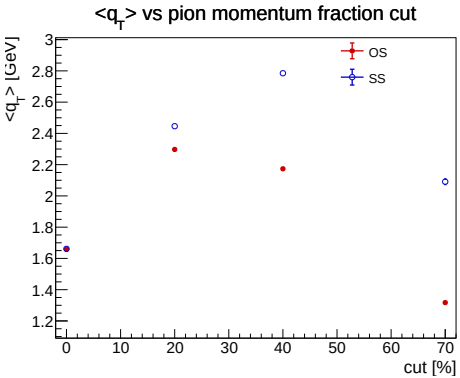
Fragmentation-function ratio vs z



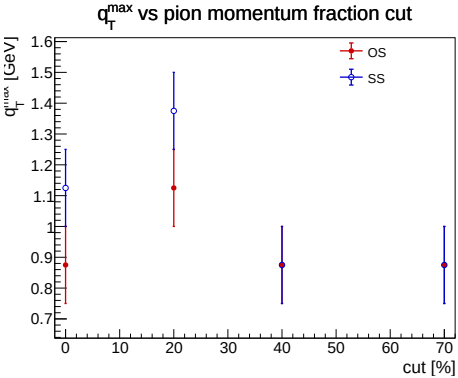
OS/SS ratio vs pion momentum-fraction cut

As the momentum fraction increases, the favoured / unfavoured fragmentation ratio grows, which is consistent with the stronger OS/SS separation at harder cuts.

Cut-dependent observables



Mean q_T vs cut



$q_T^{\max}$ vs cut

For $\langle q_T \rangle$, OS and SS start close together but SS becomes clearly larger at harder cuts, especially at 40% and 70%, suggesting the SS distribution stays harder. For $q_T^{\max}$, the OS and SS peak positions remain much closer, so the main difference appears to come more from the overall shape and tail than from a large shift in the peak.

Main conclusions

- The updated TMD theory input now provides a useful external shape comparison in the small-q_T region.
- Favoured vs unfavoured fragmentation offers a plausible qualitative explanation for the increasing OS/SS separation at larger cut.
- Cut-dependent observables such as OS/SS, $\langle q_T \rangle$, and $q_T^{\max}$ provide a more compact way to track the difference between OS and SS production.

Questions?

Starting point of the theory calculation

Input

The calculation starts from ordinary collinear pion fragmentation functions from DSS14:

$$D_i^{\pi^+}(z, Q).$$

- These depend only on the hadron momentum fraction z and the hard scale Q.
- They do **not** yet contain explicit transverse-momentum dependence.
- APFEL++ then builds **TMD fragmentation functions** from these inputs.

$$D(z, Q) \longrightarrow D(z, b; \mu, \zeta)$$

The TMD FFs are constructed in b-space, and the calculation is done at fixed

$$z_1 = z_2 = 0.7, \qquad Q = M_Z.$$

OS and SS luminosities

Opposite-sign combination

$$\mathcal{L}_{\mathrm{OS}}(b) = \sum_q B_q \left[D_q^{\pi^+}(z_1, b)\, D_{\bar{q}}^{\pi^+}(z_2, b) + D_{\bar{q}}^{\pi^+}(z_1, b)\, D_q^{\pi^+}(z_2, b) \right].$$

Same-sign combination

$$\mathcal{L}_{\mathrm{SS}}(b) = \sum_q B_q \left[D_q^{\pi^+}(z_1, b)\, D_{\bar{q}}^{\pi^+}(z_2, b) + D_{\bar{q}}^{\pi^+}(z_1, b)\, D_q^{\pi^+}(z_2, b) \right].$$

The difference between OS and SS comes from how the quark and antiquark fragmentation functions are paired, with each flavour weighted by the electroweak factor B_q.

Final outputs of the calculation

From b-space to q_T

The b-space luminosities are numerically transformed into the final q_T-dependent theory curves:

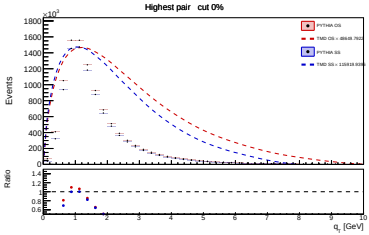
$$\mathcal{L}_{\mathrm{OS,SS}}(b) \longrightarrow \frac{d\sigma_{\mathrm{OS,SS}}}{dq_T}.$$

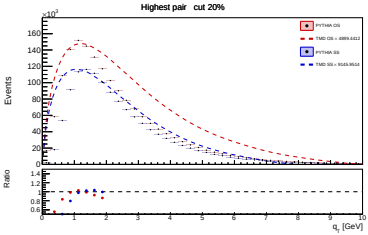
- This gives the OS and SS theory curves that are overlaid with the Pythia results.
- The code also prints the ratio

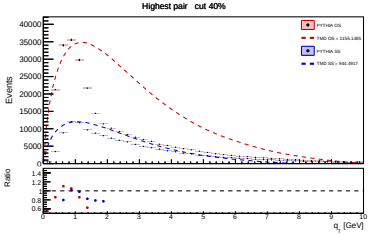
$$\frac{D_u^{\pi^+}(z,Q)}{D_{\bar{u}}^{\pi^+}(z,Q)},$$

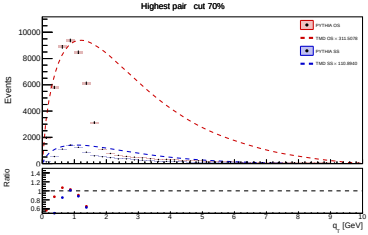
  as a function of z.
- This acts as a proxy for favoured vs unfavoured fragmentation.
- If this ratio is well above 1 at large z, harder cuts should enhance the OS/SS separation.

Event shape comparisons

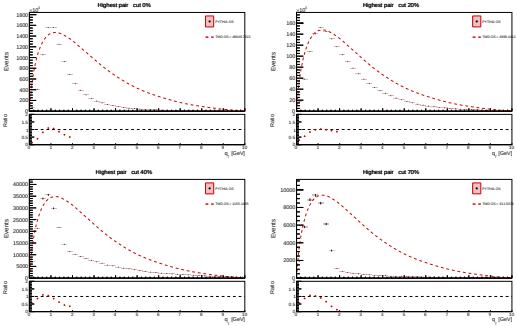




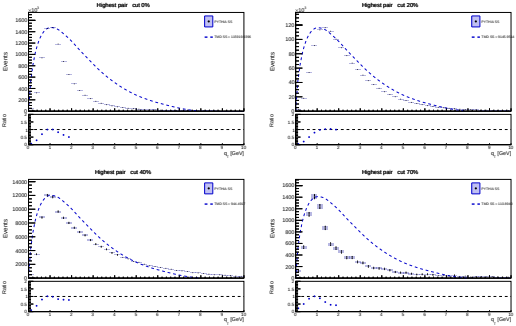




OS and SS event examples



OS event



SS event